

GGR214 Practical 3: Atmospheric Moisture Conditions

Part A

Q1

	Year	DOY	Hour	Ta (C)	Ts (C)	RH (%)	wind_speed (ms-1)	wind_dir (Deg)	precip (mm)	es (kPa)
0	2013	1	0:30	-1.24	-1.614355	75.90	6.6	256.7	0.0	0.558110
1	2013	1	1:00	-1.40	-1.990295	72.50	6.3	281.8	0.0	0.551595
2	2013	1	1:30	-1.81	-2.547568	73.80	6.1	262.0	0.0	0.535210
3	2013	1	2:00	-2.25	-3.531440	73.30	5.4	275.1	0.0	0.518115
4	2013	1	2:30	-2.45	-3.668000	69.64	5.2	269.3	0.0	0.510508
...	...	...	...	...	...	...	...	...	...	...
139	2013	213	22:00	15.33	16.488227	96.00	1.7	276.2	0.0	1.754800
140	2013	213	22:30	14.96	16.276745	97.30	1.6	285.5	0.0	1.712946
141	2013	213	23:00	14.91	15.957257	98.60	1.7	283.1	0.0	1.707359
142	2013	213	23:30	14.69	15.388115	99.50	1.2	296.0	0.0	1.682968
143	2013	213	0:00	14.32	15.071605	100.00	1.4	287.6	0.0	1.642650

144 rows × 10 columns

Q2

	Year	DOY	Hour	Ta (C)	Ts (C)	RH (%)	wind_speed (ms-1)	wind_dir (Deg)	precip (mm)	es (kPa)	e (kPa)
0	2013	1	0:30	-1.24	-1.614355	75.90	6.6	256.7	0.0	0.558110	0.423606
1	2013	1	1:00	-1.40	-1.990295	72.50	6.3	281.8	0.0	0.551595	0.399906
2	2013	1	1:30	-1.81	-2.547568	73.80	6.1	262.0	0.0	0.535210	0.394985
3	2013	1	2:00	-2.25	-3.531440	73.30	5.4	275.1	0.0	0.518115	0.379778
4	2013	1	2:30	-2.45	-3.668000	69.64	5.2	269.3	0.0	0.510508	0.355518
...	...	...	...	...	...	...	...	...	...	...	...
139	2013	213	22:00	15.33	16.488227	96.00	1.7	276.2	0.0	1.754800	1.684608
140	2013	213	22:30	14.96	16.276745	97.30	1.6	285.5	0.0	1.712946	1.666696
141	2013	213	23:00	14.91	15.957257	98.60	1.7	283.1	0.0	1.707359	1.683456
142	2013	213	23:30	14.69	15.388115	99.50	1.2	296.0	0.0	1.682968	1.674554
143	2013	213	0:00	14.32	15.071605	100.00	1.4	287.6	0.0	1.642650	1.642650

144 rows × 11 columns

Q3

	Year	DOY	Hour	Ta (C)	Ts (C)	RH (%)	wind_speed (ms-1)	wind_dir (Deg)	precip (mm)	es (kPa)	e (kPa)	VPD (kPa)
0	2013	1	0:30	-1.24	-1.614355	75.90	6.6	256.7	0.0	0.558110	0.423606	0.134505
1	2013	1	1:00	-1.40	-1.990295	72.50	6.3	281.8	0.0	0.551595	0.399906	0.151689
2	2013	1	1:30	-1.81	-2.547568	73.80	6.1	262.0	0.0	0.535210	0.394985	0.140225
3	2013	1	2:00	-2.25	-3.531440	73.30	5.4	275.1	0.0	0.518115	0.379778	0.138337
4	2013	1	2:30	-2.45	-3.668000	69.64	5.2	269.3	0.0	0.510508	0.355518	0.154990
...	...	...	...	...	...	...	...	...	...	...	...	...
139	2013	213	22:00	15.33	16.488227	96.00	1.7	276.2	0.0	1.754800	1.684608	0.070192
140	2013	213	22:30	14.96	16.276745	97.30	1.6	285.5	0.0	1.712946	1.666696	0.046250
141	2013	213	23:00	14.91	15.957257	98.60	1.7	283.1	0.0	1.707359	1.683456	0.023903
142	2013	213	23:30	14.69	15.388115	99.50	1.2	296.0	0.0	1.682968	1.674554	0.008415
143	2013	213	0:00	14.32	15.071605	100.00	1.4	287.6	0.0	1.642650	1.642650	0.000000

Q4

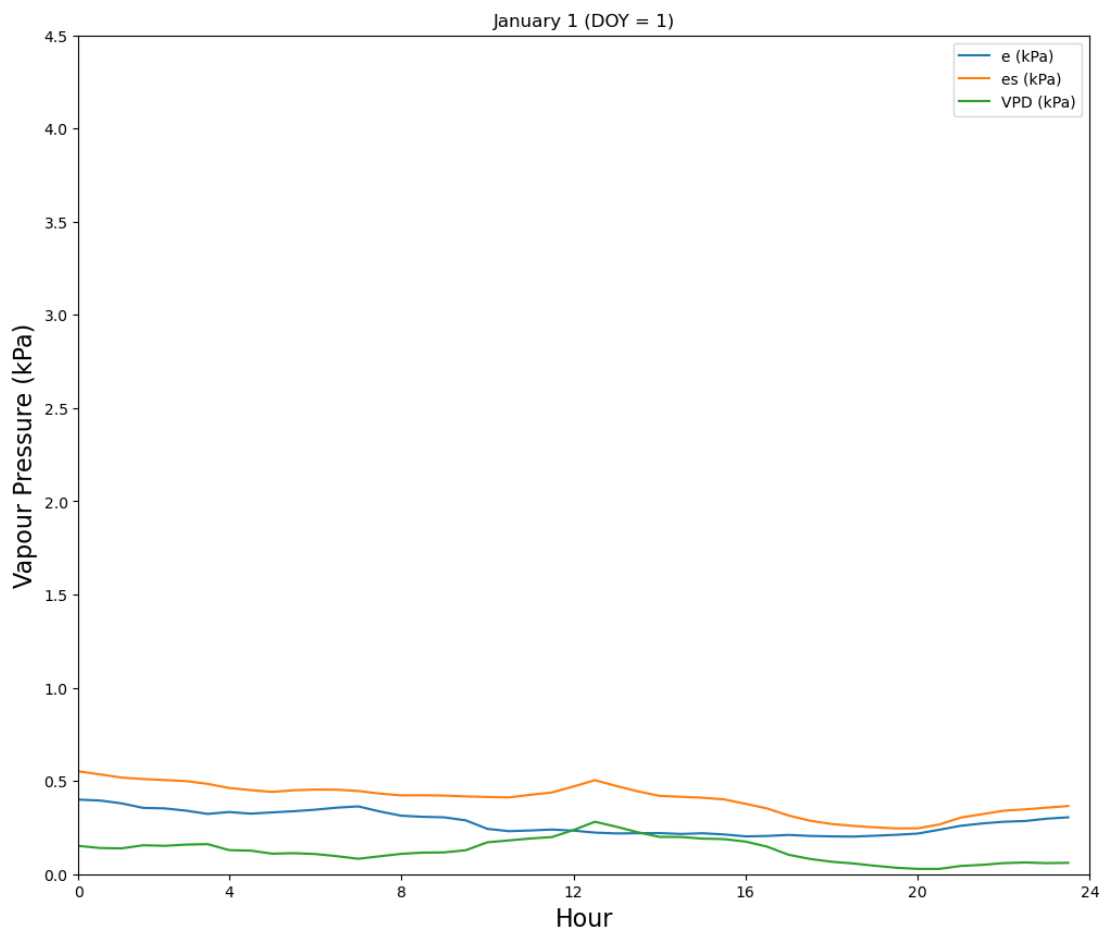


Figure 1 (a)

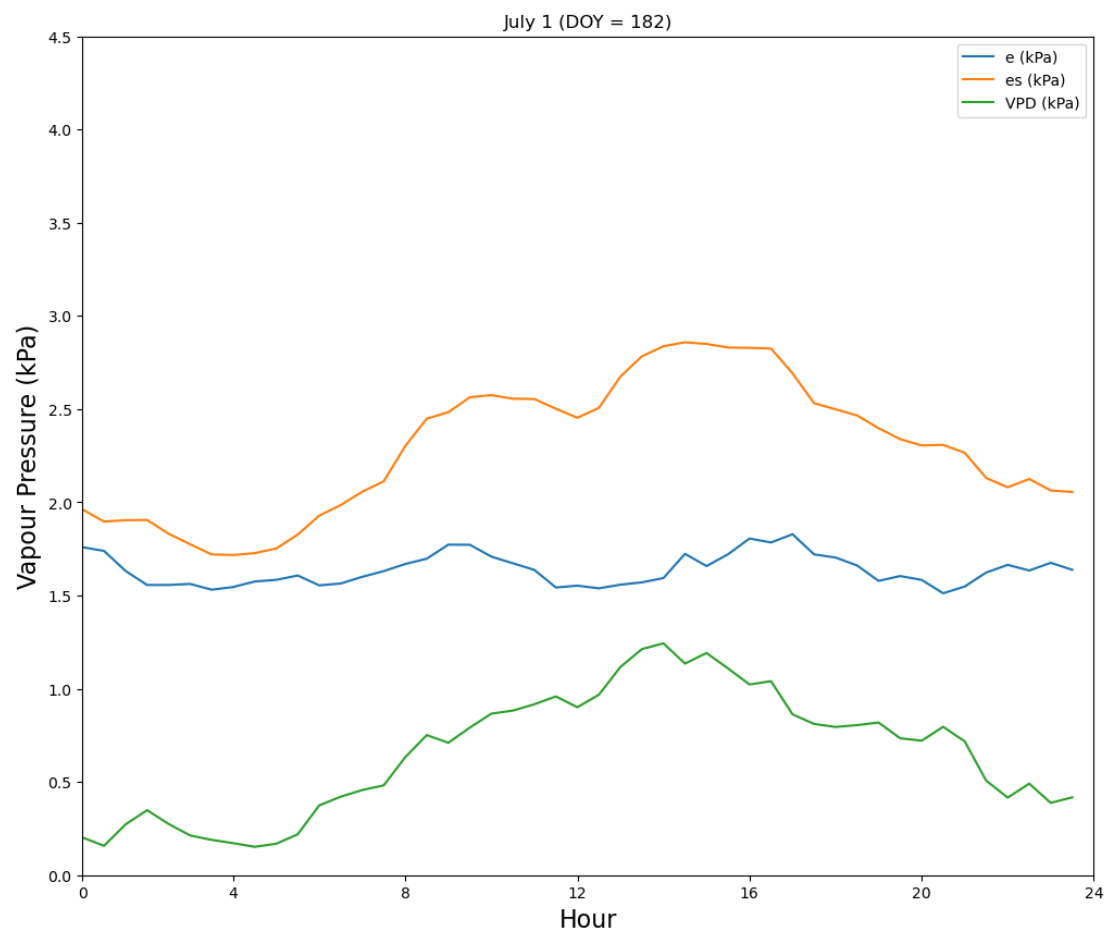


Figure 1 (b)

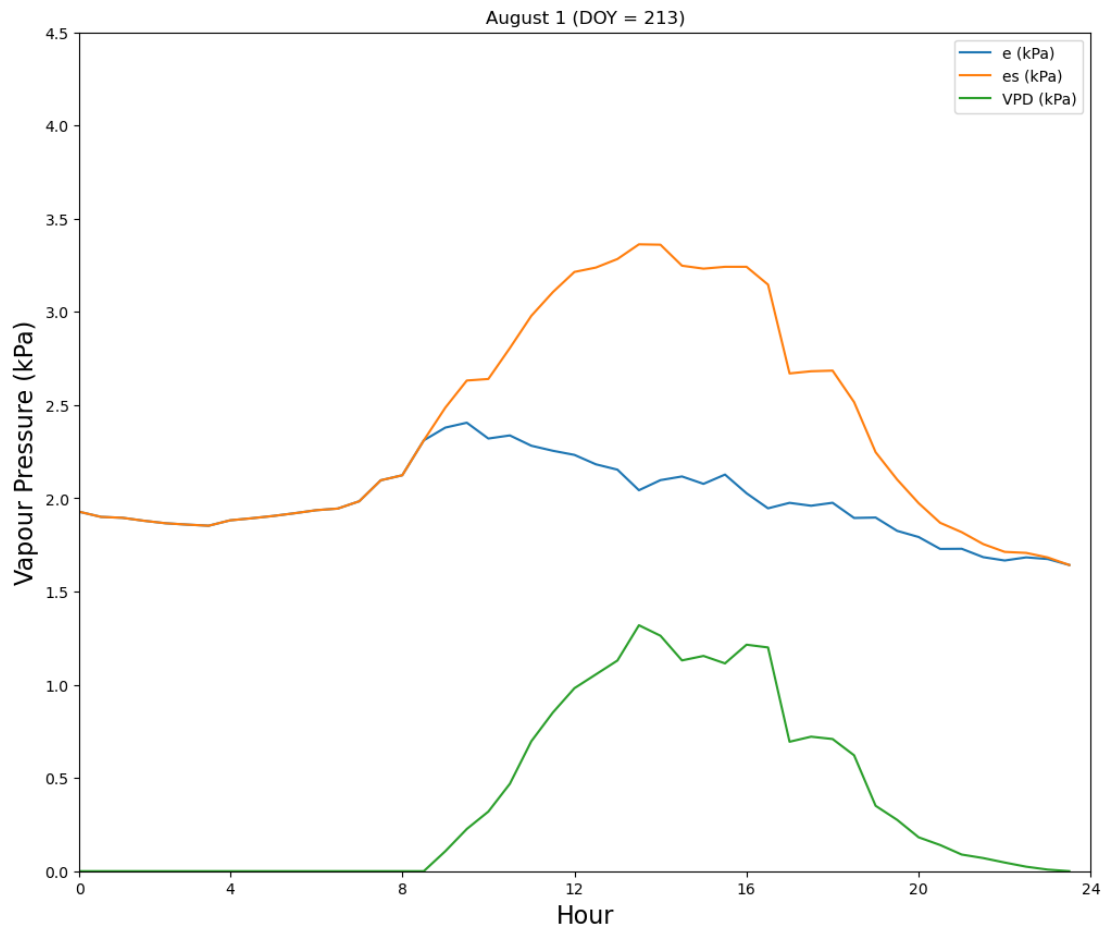


Figure 1 (c)

Figure 1 (a) (b) (c). Diurnal variations of es, e, and VPD (kPa) versus hour for January 1 (a), July 1 (b), and August 1 (c).

Q5

a) Saturated Vapor Pressure (es)

The saturated vapor pressure (es) showed significant variations across the three dates. On January 1st, es remained low throughout the day, ranging from approximately 0.4 to 0.55 kPa. July 1st saw higher es values, rising from about 1.8 kPa in the early morning to nearly 2.8 kPa in the afternoon, before declining in the evening. August 1st had the highest es values, peaking in the afternoon at approximately 3.3 to 3.4 kPa. The es values on these three days gradually increased from morning to afternoon and gradually decreased in the evening, likely related to intraday temperatures. Overall, daytime es were generally higher than nighttime es, and summer es were higher than winter es.

b) Vapor Pressure (e)

On January 1st, e is relatively stable, roughly between 0.25 and 0.4 kPa, decreasing in the early morning and rising slightly in the afternoon. This is because the water vapor content in the cold air is low and does not change much throughout the day. On July 1st, the e value in the early morning is about 1.7 kPa, then gradually increases, reaching about 1.8 kPa in the afternoon, and then decreases again in the evening. On the morning of August 1st, the e value is about 1.9 kPa, then steadily increases, reaching a peak of about 2.3–2.4 kPa at noon, and then gradually decreases to about 1.6–1.7 kPa in the afternoon and evening. Overall, e also increases during the day and decreases in the evening, and is lowest in winter and higher in summer.

c) Vapor Pressure Difference (VPD)

The diurnal variation of vapor pressure difference (VPD) is most pronounced. On the morning of August 1st, VPD is approximately 0 kPa, indicating near-saturation ($e \approx e_s$). It peaks in the afternoon at approximately 1.2–1.3 kPa, coinciding with the highest air temperature and lowest relative humidity. In the evening, as temperatures drop, VPD decreases and approaches zero.

On July 1st, VPD also increases over time, starting at a relatively high value of approximately 0.2–0.3 kPa and peaking at approximately 1.2 kPa, indicating low relative humidity during the day. On January 1st, VPD values are very low throughout the day, typically below 0.2 kPa, indicating near-saturation.

The reason VPD is 0 on August 1st is due to the low morning temperature and high humidity, leading to near-saturation. As solar radiation intensifies during the day, temperatures rise, causing e_s to increase rapidly, while e changes more slowly. This means the difference between e and e_s widens, resulting in an increase in VPD. At night, the temperature drops, e_s decreases, relative humidity increases, and VPD decreases.

Part B

Q6

	Year	DOY	Hour	Ta (C)	Ts (C)	RH (%)	wind_speed (ms-1)	wind_dir (Deg)	precip (mm)	es (kPa)	e (kPa)	VPD (kPa)	Td (C)
0	2013	1	0:30	-1.24	-1.614355	75.90	6.6	256.7	0.0	0.558110	0.423606	0.134505	-6.060
1	2013	1	1:00	-1.40	-1.990295	72.50	6.3	281.8	0.0	0.551595	0.399906	0.151689	-6.900
2	2013	1	1:30	-1.81	-2.547568	73.80	6.1	262.0	0.0	0.535210	0.394985	0.140225	-7.050
3	2013	1	2:00	-2.25	-3.531440	73.30	5.4	275.1	0.0	0.518115	0.379778	0.138337	-7.590
4	2013	1	2:30	-2.45	-3.668000	69.64	5.2	269.3	0.0	0.510508	0.355518	0.154990	-8.522
...	...	...	...	...	...	...	...	...	...	...	...	...	...
139	2013	213	22:00	15.33	16.488227	96.00	1.7	276.2	0.0	1.754800	1.684608	0.070192	14.530
140	2013	213	22:30	14.96	16.276745	97.30	1.6	285.5	0.0	1.712946	1.666696	0.046250	14.420
141	2013	213	23:00	14.91	15.957257	98.60	1.7	283.1	0.0	1.707359	1.683456	0.023903	14.630
142	2013	213	23:30	14.69	15.388115	99.50	1.2	296.0	0.0	1.682968	1.674554	0.008415	14.590
143	2013	213	0:00	14.32	15.071605	100.00	1.4	287.6	0.0	1.642650	1.642650	0.000000	14.320

144 rows × 13 columns

Q7

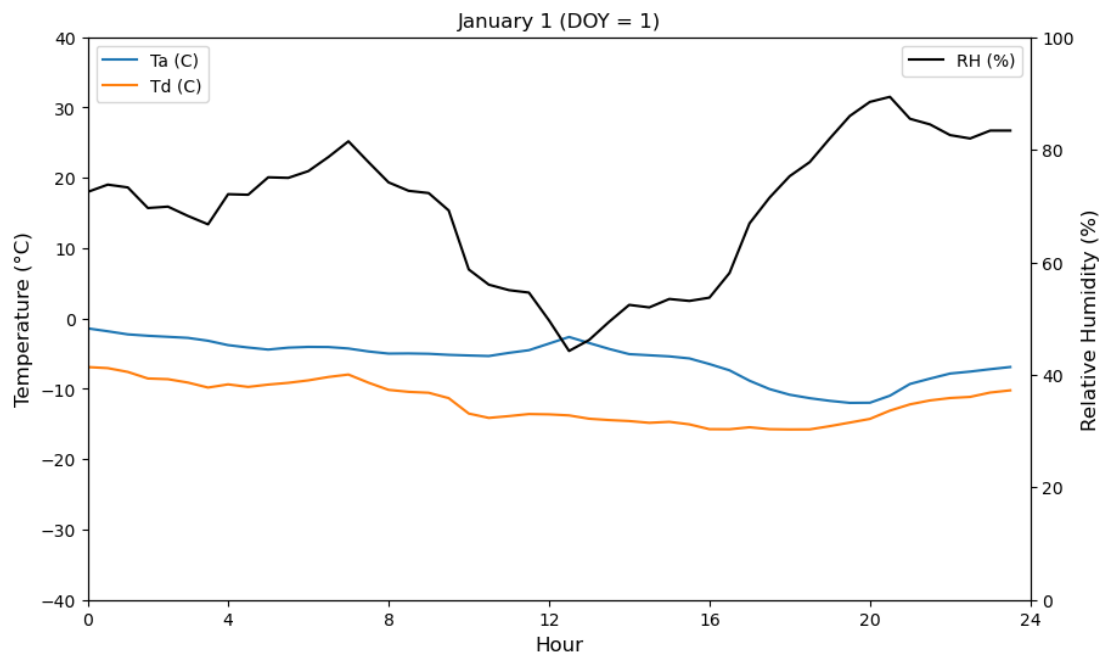


Figure 2 (a)

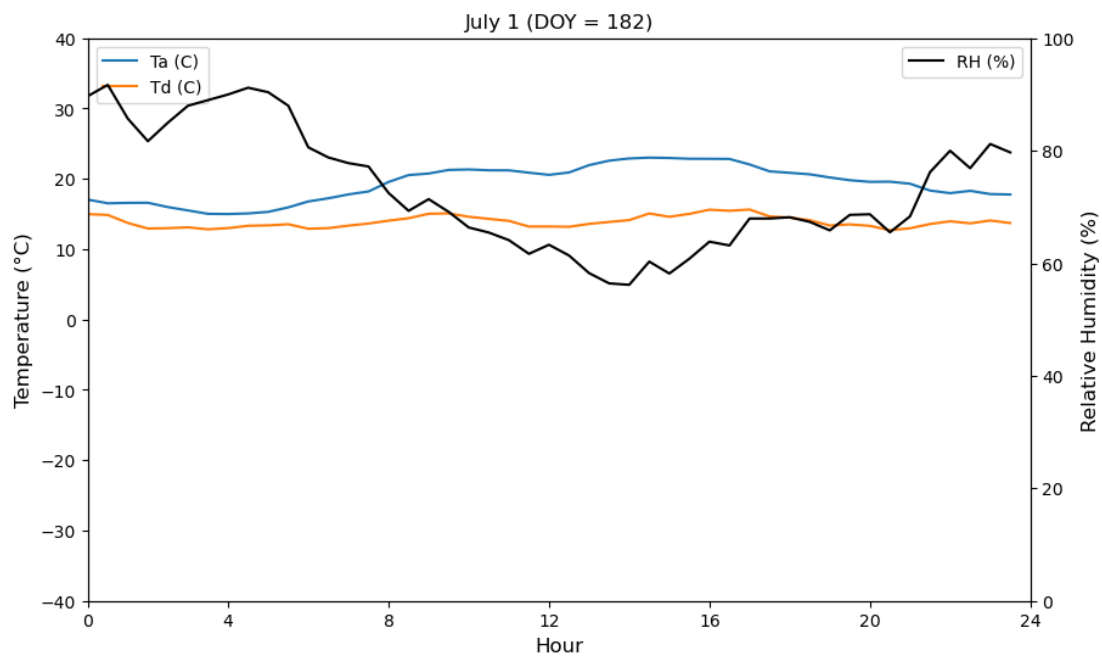


Figure 2 (b)

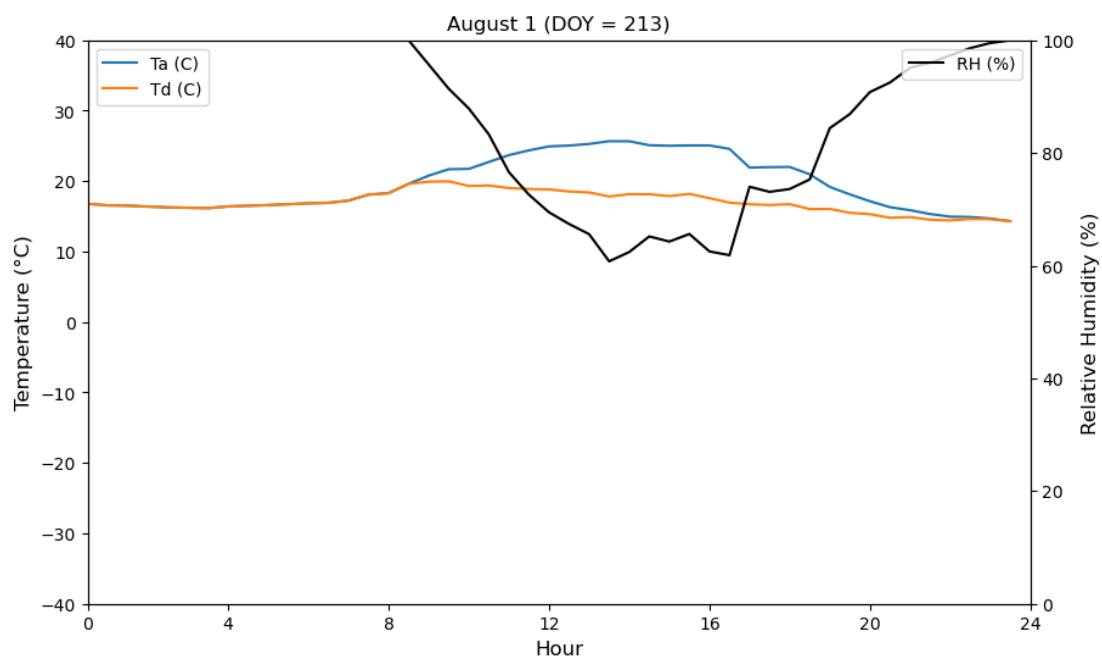


Figure 2 (c)

Figure 2. Diurnal variations of air temperature (Ta), dewpoint temperature (Td), and relative humidity (RH) versus hour for January 1 (a), July 1 (b), and August 1 (c) at the UTM weather station.

Q8

a)

On August 1st, the temperature (T_a) exhibited a clear diurnal variation. The temperature was low in the early morning, gradually rising after sunrise, reaching a peak of approximately 25–26°C in the afternoon, and then gradually decreasing in the evening. The dew point temperature (T_d) showed a smaller diurnal variation, typically between 15 and 20°C, rising slightly in the morning and then gradually decreasing. The relative humidity (RH) varied in the opposite direction to the temperature; it was very high in the early morning, approaching 100%, then dropped rapidly in the morning, reaching only about 10% in the afternoon, before rising again to near saturation in the evening.

As daytime temperatures rise, the air's ability to hold water vapor increases. Therefore, even if the actual humidity does not change significantly, the relative humidity will decrease. This explains the significant negative correlation between temperature (T_a) and relative humidity (RH) on August 1st. The change in dew point temperature (T_d) is much smaller than that of temperature because dew point primarily reflects the actual humidity of the air, while temperature reflects the heating effect of solar radiation during the day.

b)

In Figure 1c, VPD is 0 in the early morning, then rises rapidly, peaking in the afternoon at approximately 1.2–1.3 kPa, and then declines again in the evening.

In the early morning, relative humidity is close to 100%, and T_a and T_d are very close. At this time, the air is nearly saturated, e is almost equal to e_s , so VPD is close to zero. As time progresses, T_a increases significantly, and e_s increases because saturated vapor pressure is temperature-dependent. The small change in T_d means that e increases less rapidly than e_s . RH drops sharply, widening the gap between e_s and e . Therefore, VPD increases in the late morning and afternoon.

In the evening, T_a decreases, RH rises again, while T_d remains relatively stable or decreases only slightly. At this time, e_s decreases, narrowing the gap between e_s and e , thus causing VPD to drop back to near zero. Therefore, the temperature and humidity changes shown in Figure 2c provide an explanation for the water vapor pressure change pattern on August 1st shown in Figure 1c.

Q9

	Year	DOY	Hour	Ta (C)	Ts (C)	RH (%)	wind_speed (ms-1)	wind_dir (Deg)	precip (mm)	es (kPa)	e (kPa)	VPD (kPa)	Td (C)	LCL (m)
0	2013	1	0:30	-1.24	-1.614355	75.90	6.6	256.7	0.0	0.558110	0.423606	0.134505	-6.060	555.705597
1	2013	1	1:00	-1.40	-1.990295	72.50	6.3	281.8	0.0	0.551595	0.399906	0.151689	-6.900	613.713136
2	2013	1	1:30	-1.81	-2.547568	73.80	6.1	262.0	0.0	0.535210	0.394985	0.140225	-7.050	562.804035
3	2013	1	2:00	-2.25	-3.531440	73.30	5.4	275.1	0.0	0.518115	0.379778	0.138337	-7.590	507.320008
4	2013	1	2:30	-2.45	-3.668000	69.64	5.2	269.3	0.0	0.510508	0.355518	0.154990	-8.522	606.749975
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
139	2013	213	22:00	15.33	16.488227	96.00	1.7	276.2	0.0	1.754800	1.684608	0.070192	14.530	244.778404
140	2013	213	22:30	14.96	16.276745	97.30	1.6	285.5	0.0	1.712946	1.666696	0.046250	14.420	232.093137
141	2013	213	23:00	14.91	15.957257	98.60	1.7	283.1	0.0	1.707359	1.683456	0.023903	14.630	165.907110
142	2013	213	23:30	14.69	15.388115	99.50	1.2	296.0	0.0	1.682968	1.674554	0.008415	14.590	99.764383
143	2013	213	0:00	14.32	15.071605	100.00	1.4	287.6	0.0	1.642650	1.642650	0.000000	14.320	93.950675

144 rows × 14 columns

Q10

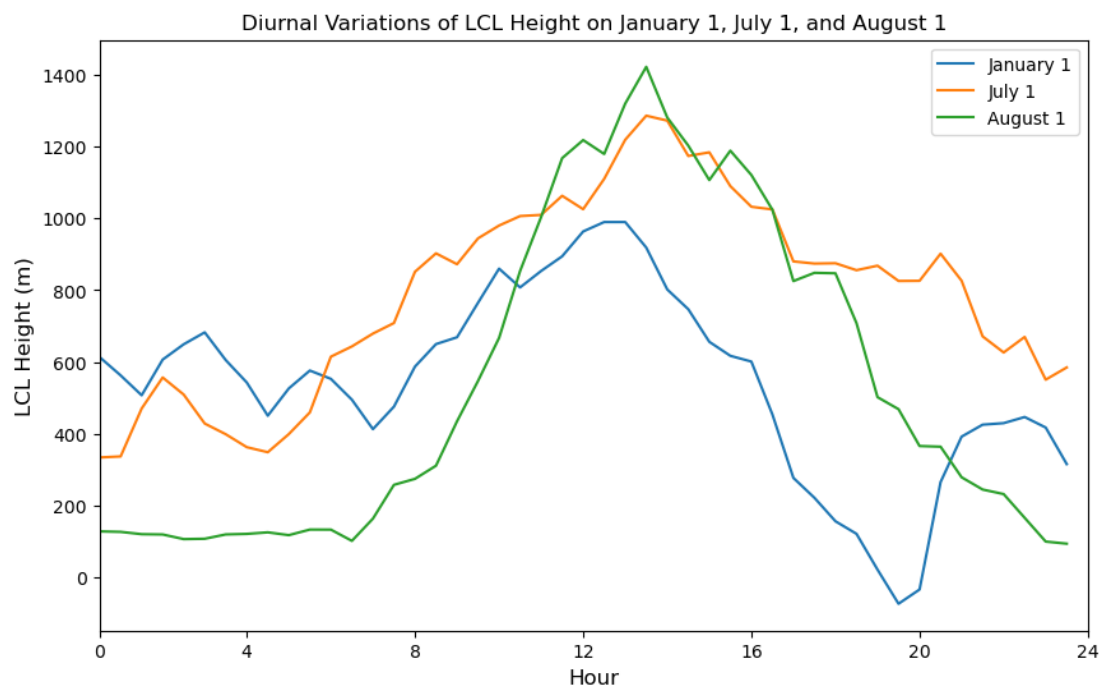


Figure 3 Diurnal variations of LCL height (m) versus hour for January 1, July 1, and August 1 at the UTM weather station.

Q11

a)

According to the LCL chart, the highest LCL peak was on August 1st, reaching approximately 1400-1450 meters in the early afternoon; the lowest LCL peak was on January 1st, reaching approximately 950-1000 meters; the LCL peak on July 1st was between the two, at approximately 1250-1300 meters.

b)

Clouds could form below 400 meters on both August 1st and January 1st, but the possibility is lower on July 1st.

On the morning of August 1st, the LCL (Cloud Base Level) was very low, around 100-150 meters, indicating that condensation could occur at extremely low altitudes under near-saturation conditions. On January 1st, the cloud base occasionally approached 400 meters, potentially allowing for the formation of low-level clouds at certain times. On July 1st, the cloud base remained almost entirely above 400 meters, making it unlikely that low-level clouds would form below 400 meters.

c)

On the morning of August 1st, the cloud base (LCL) was very low (approximately 100-150 meters), with high relative humidity, which favored the formation of stratocumulus or fog. As time progressed, the cloud base rose above 1400 meters, and surface heating exacerbated atmospheric instability, likely indicating the development of stratocumulus. Looking at the precipitation map for August 1st, precipitation was observed in the early morning, making stratocumulus the most probable cloud type.

On July 1st, the cloud base remained consistently high. Combined with warm temperatures and moderate humidity, these conditions were characteristic of clear cumulus. The high LCL suggests that cumulus or fog was unlikely. Looking at the precipitation and radiation charts, less radiation was observed at midday compared to other times, likely due to thick cloud cover. Combined with the lack of precipitation during this period, the clouds at this time were confirmed to be cumulus.

Part C

Q12

	DOY	Ta (C)	Tmax (C)	RH (%)	wind_speed (ms-1)	wind_dir (Deg)	precip (mm)	es (kPa)	e (kPa)	Td (C)
0	1	-5.73	-4.0	69.63	4.18	254.28	0.0	0.399289	0.278025	-11.804
1	2	-5.35	-1.0	80.17	4.17	254.59	0.0	0.410946	0.329455	-9.316
2	3	-3.83	0.0	87.62	3.81	242.97	0.0	0.460703	0.403668	-6.306
3	4	-1.54	-0.5	78.05	7.75	248.08	0.0	0.545950	0.426114	-5.930
4	5	-0.17	3.5	83.62	4.34	228.29	0.0	0.603492	0.504640	-3.446
...	...	...	...	...	...	...	...	...	...	...
360	361	-3.76	3.0	98.75	0.19	269.56	0.0	0.463120	0.457331	-4.010
361	362	1.10	5.0	101.21	2.74	240.34	0.0	0.661645	0.669651	1.342
362	363	1.57	5.0	89.60	4.66	249.72	0.0	0.684412	0.613233	-0.510
363	364	-9.02	-7.0	69.46	4.45	241.92	0.0	0.310157	0.215435	-15.128
364	365	-8.53	-6.0	81.18	4.87	240.14	0.0	0.322176	0.261543	-12.294

365 rows × 10 columns

Q13

	DOY	Ta (C)	Tmax (C)	RH (%)	wind_speed (ms-1)	wind_dir (Deg)	precip (mm)	es (kPa)	e (kPa)	Td (C)	humidex
0	1	-5.73	-4.0	69.63	4.18	254.28	0.0	0.399289	0.278025	-11.804	-8.010974
1	2	-5.35	-1.0	80.17	4.17	254.59	0.0	0.410946	0.329455	-9.316	-4.725249
2	3	-3.83	0.0	87.62	3.81	242.97	0.0	0.460703	0.403668	-6.306	-3.312956
3	4	-1.54	-0.5	78.05	7.75	248.08	0.0	0.545950	0.426114	-5.930	-3.688255
4	5	-0.17	3.5	83.62	4.34	228.29	0.0	0.603492	0.504640	-3.446	0.748000
...	...	...	...	...	...	...	...	...	...	...	...
360	361	-3.76	3.0	98.75	0.19	269.56	0.0	0.463120	0.457331	-4.010	-0.014829
361	362	1.10	5.0	101.21	2.74	240.34	0.0	0.661645	0.669651	1.342	3.164728
362	363	1.57	5.0	89.60	4.66	249.72	0.0	0.684412	0.613233	-0.510	2.851293
363	364	-9.02	-7.0	69.46	4.45	241.92	0.0	0.310157	0.215435	-15.128	-11.358694
364	365	-8.53	-6.0	81.18	4.87	240.14	0.0	0.322176	0.261543	-12.294	-10.102540

365 rows × 11 columns

Q14

a)

Based on the calculated humidex values, there were 79 days with a humidex exceeding 29°C, 15 days humidex exceeds 40, and 29 days with a daily high temperature exceeding 29°C. The number of days with a humidex exceeding 29°C is higher than the number of days with an actual temperature exceeding 29°C. This is

likely due to humidity; even if the actual temperature is not high, high humidity can increase the humidex.

b)

There are 46 days a year when the dew point temperature exceeds 18°C. On these days, the air humidity is high, and the body's ability to dissipate heat through evaporation decreases, which may cause discomfort.

c)

The number of days identified using humidex and dewpoint temperature differs. Humidex considers both the current humidity and air temperature. Dew point temperature measures the water vapor content in the air and is independent of temperature. A higher dew point value indicates higher air humidity, which reduces evaporative heat dissipation and makes people feel uncomfortable. Therefore, dew point temperature can be considered a simpler indicator to measure a person's comfort level in an environment.